

Chapter 8: Communication Systems

Department of Electrical & Electronic Engineering

2021-Fall

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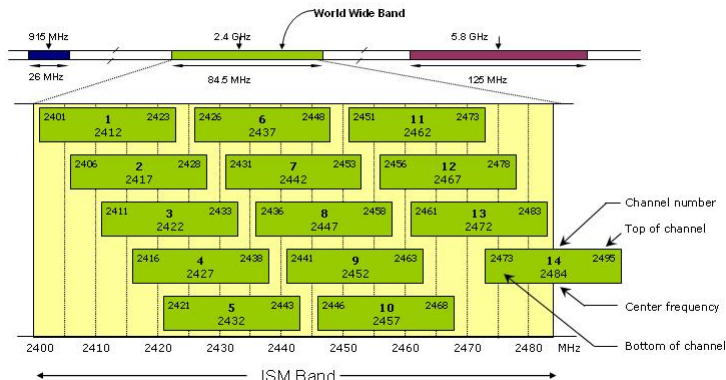


Why Modulation?

- **Modulation:** Embed an information-bearing signal into a second signal.

$$\text{E.g., } y(t) = x(t)\cos\omega t$$

- **Example 1:** Voice range $\sim [200\text{Hz}, 4\text{kHz}]$, microwave link $\sim [300\text{MHz}, 300\text{GHz}]$
- **Example 2:** Wifi in industrial, scientific and medical (ISM) radio bands



Amplitude Modulation

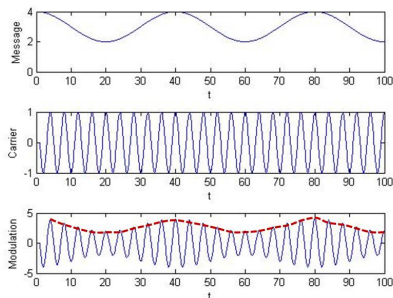
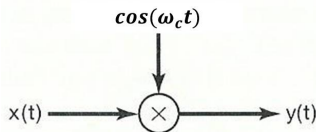
- General form

$$y(t) = x(t)c(t)$$

- $x(t)$: modulating signal with information
- $c(t)$: carrier signal
- Complex exponential carrier: $c(t) = e^{j\omega_c t + \theta_c}$
- Sinusoidal carrier: $c(t) = \cos(\omega_c t + \theta_c)$
- **Question**: how to generate complex exponential carrier?



AM with a Sinusoidal Carrier

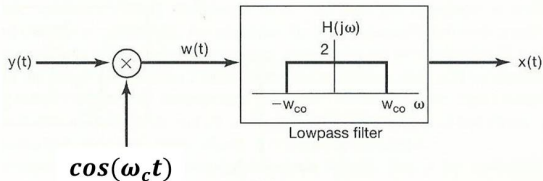


- Time & frequency domain expression

$$y(t) = x(t)\cos(\omega_c t) \longleftrightarrow Y(j\omega) = \frac{1}{2}[X(j\omega - j\omega_c) + X(j\omega + j\omega_c)]$$

Synchronous Demodulation

Sinusoidal Carrier:



- Requirement: oscillators at the transmitters and receivers should be synchronized
- What happen if it's not in phase?
- What happen if frequency offset exists?

Phase Error

The carrier phase of the receiver may not synchronize with the transmitter at the very beginning.

$$\begin{aligned}w(t) &= x(t)\cos(\omega_c t)\cos(\omega_c t + \Delta) \\ &= \frac{1}{2}x(t) [\cos(2\omega_c t + \Delta) + \cos(\Delta)]\end{aligned}$$

After lowpass filter

$$\hat{x}(t) = x(t)\cos(\Delta)$$

Frequency Offset

The carrier frequency at the receiver may not synchronize with the transmitter.

$$\begin{aligned}w(t) &= x(t)\cos(\omega_c t)\cos(\omega_c t + \Delta t) \\&= \frac{1}{2}x(t)[\cos(2\omega_c t + \Delta t) + \cos(\Delta t)]\end{aligned}$$

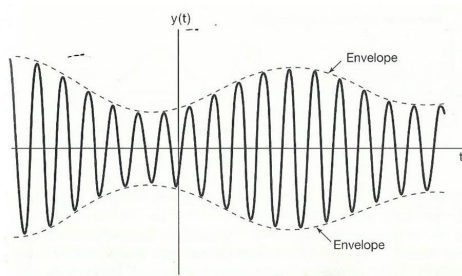
After lowpass filter

$$\hat{x}(t) = x(t)\cos(\Delta t)$$

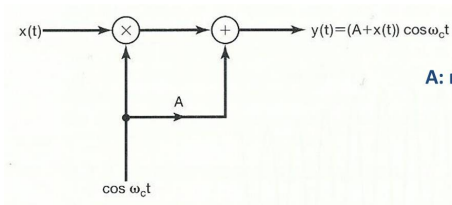
Modulator for Asynchronous Demodulation

- AM signal is like

Envelope of the carrier
taking information:
(Negative Signal??)



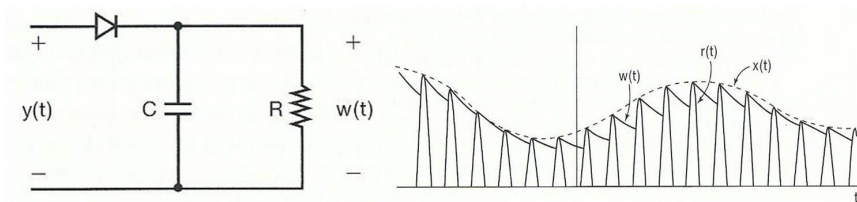
- Modulator structure (Percent Modulation)



A: make sure $A+x(t)$ is positive



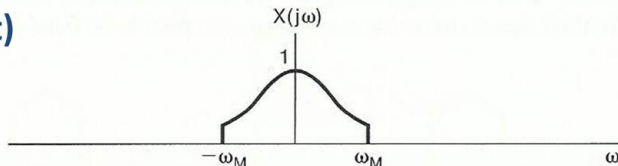
Envelope Detector



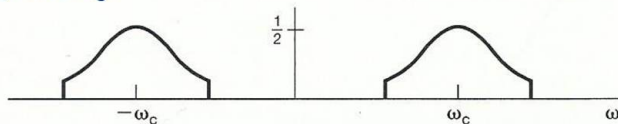
- Requirement: Carrier frequency $\gg$ Signal bandwidth
- Larger transmission power v.s. simpler receiver (asynchronous)
 - ▶ Public radio broadcasting
- Question
 - ▶ How to improve the quality of $w(t)$?

AM Spectrum s/w Carrier

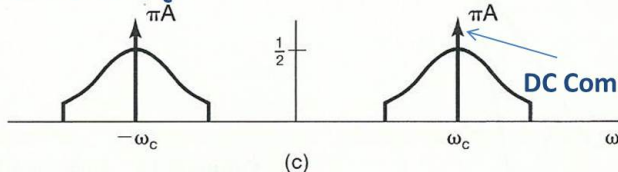
$x(t)$



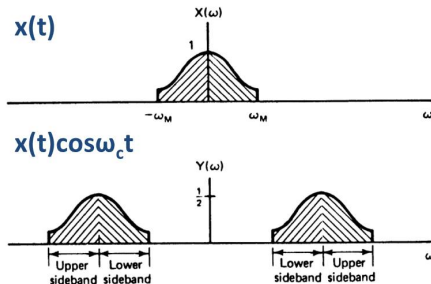
$x(t)\cos\omega_c t$



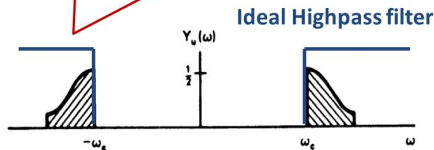
$[x(t)+A]\cos\omega_c t$



Single-Sideband AM

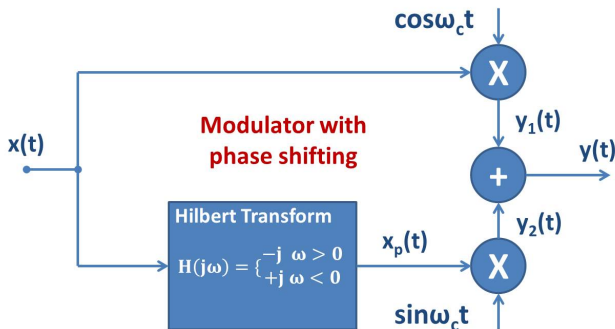


Higher power and
spectrum efficiency



- **Observation:** each sideband (**lower sideband** or **upper sideband**) contains the complete signal information
- Use ideal highpass/lowpass filter to obtain the upper/lower sideband
- **Question:** how to do demodulation?
- **Advantage:** save the spectrum and power

Single-Sideband AM: Another Modulator



- Questions :

- ▶ What's the output signal?
- ▶ How to filter out the other sideband?

Sinusoidal AM: Summary

- Four types of sinusoidal AM
 - ▶ AM-DSB/SC: $y(t) = x(t)\cos\omega_c t$
 - ▶ AM-DSB/WC: $y(t) = (x(t) + A)\cos\omega_c t$
 - ▶ AM-SSB/SC: AM-DSB/SC + ideal highpass/lowpass filter
 - ▶ AM-SSB/WC: AM-SSB/SC + $A\cos\omega_c t$
- Questions:
 - ▶ How to compare the transmitter/receiver complexity?
 - ▶ How to compare the efficiency?

Frequency-Division Multiplexing (FDM)

